

Synapse

A TECHNICAL STUDY OF MULTI-AGENT DOCUMENT INTELLIGENCE

Engineering answers you can trace.

From a raw document to a cited answer: the architecture, retrieval decisions, GPU pipeline, visual artifacts, and evaluation behind a multi-tenant research workspace.

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Architecture / Implementation / Evaluation

Agentic RAG / Document AI / GPU workers / Evaluation

Contents

A complete technical description of the product, its architecture, implementation boundaries, and evaluation. Select a chapter below to jump to its page.

01 / Research software should show its work.	3
02 / Three planes, different responsibilities.	4
03 / A document is more than extracted text.	6
04 / Specialists for structure; configuration for reasoning.	7
05 / Retrieval follows the question.	8
06 / A citation is part of the answer contract.	10
07 / From an answer to a reusable artifact.	11
08 / Relationships retain their provenance.	13
09 / Collaboration does not mean unrestricted access.	14
10 / Measure retrieval before judging the answer.	16
11 / The job queue is part of the product.	18
12 / A system that exposes its assumptions.	20

PART I / THE PROBLEM

01 / Research software should show its work.

Synapse addresses a familiar failure in document-based AI: an answer sounds convincing, but the reader cannot tell which passage supports it, whether a table was read correctly, or whether important evidence was missed. The product is a research workspace, not just a prompt box over a vector database.

A user connects a collection of documents, waits for structured ingestion, and asks questions across one or many files. The answer should preserve the relationship between claims, source chunks, pages, and relevant figures. Teams can collaborate over explicitly shared libraries while personal conversations remain separate.

TABLE 01 / Input / System responsibility / Output

Input	System responsibility	Output
PDFs and connected Google Drive libraries	Preserve layout, text, visuals, and document identity	Searchable, attributed evidence
A question and its conversation context	Select a retrieval strategy and test evidence coverage	A grounded answer with citations
A visual or relationship-building request	Acquire source data, validate a specification, retain provenance	Charts, diagrams, and knowledge graphs

THE ENGINEERING PRINCIPLE

The model can decide what to investigate; the runtime limits how far the investigation may go. Citations, retained artifacts, and explicit evaluation make the output inspectable, not automatically correct.

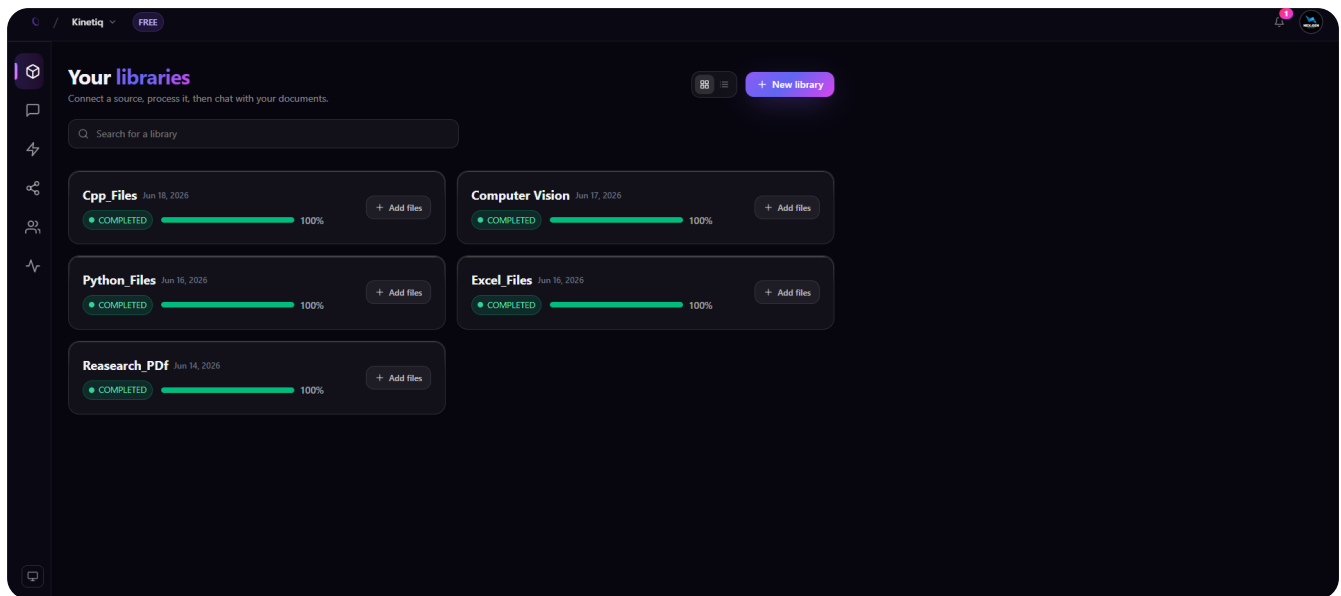


Figure 01. Libraries are the unit of connected knowledge. Processing status makes ingestion visible before chat begins.

PART II / THE SYSTEM

02 / Three planes, different responsibilities.

The system separates a responsive edge product, a GPU execution plane, and durable shared state. Next.js 16 and React 19 provide the interface. OpenNext hosts the application on Cloudflare Workers, where route handlers proxy authenticated requests, manage product state, and connect to object storage and email services.

FastAPI exposes chat, pipeline, agent, and graph services on a dedicated compute host. Ingestion is queue-backed rather than tied to a long browser request. Worker processes perform specialized stages, while the reasoning layer assembles evidence and answers. Supabase Postgres holds the tenant model, job state, conversation records, and pgvector embeddings. Cloudflare R2 holds raw and derived files.

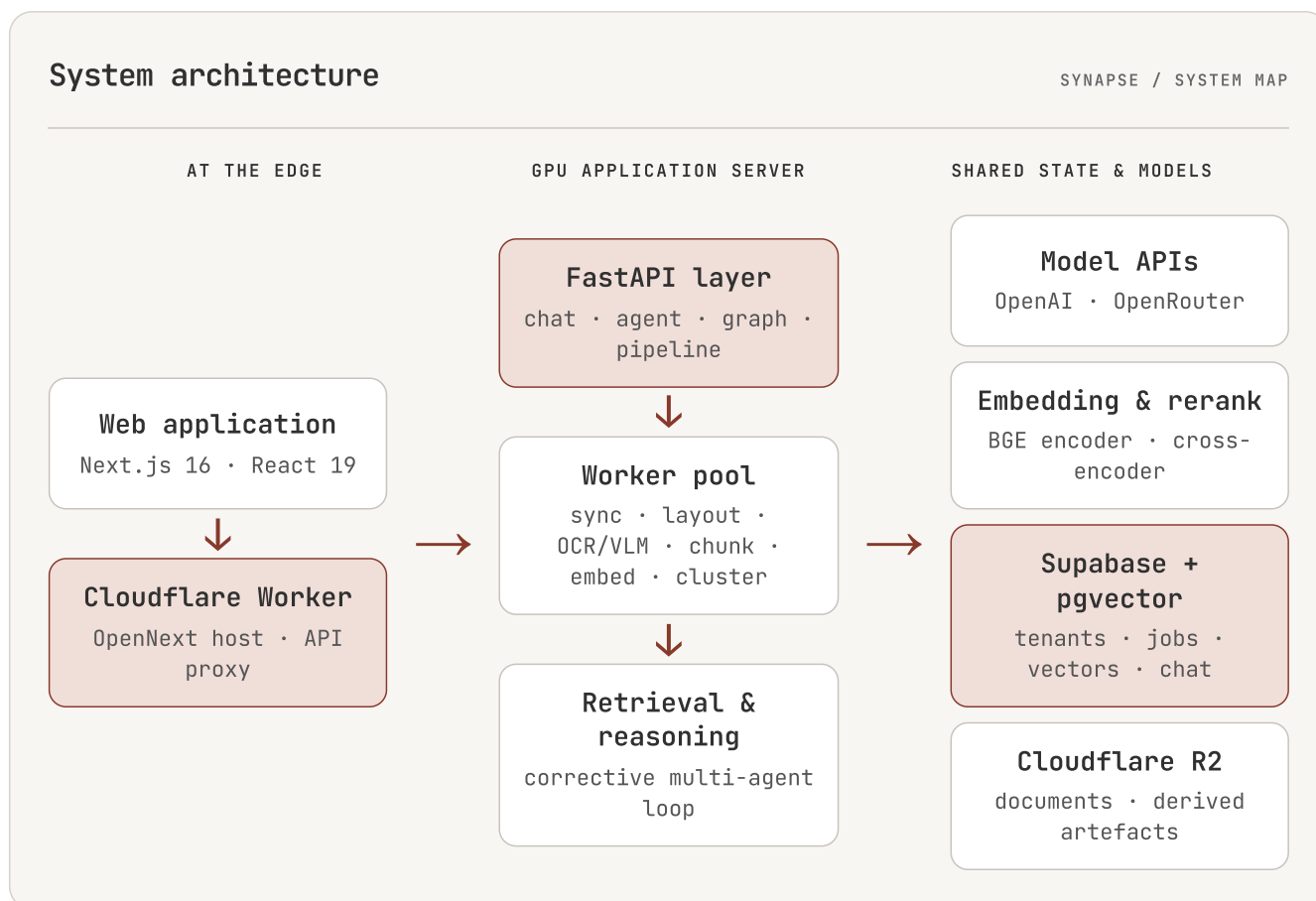


Figure 02. Next.js at the edge, a FastAPI/GPU execution plane, and durable shared state remain independently scalable.

One upload, two persistence layers

A file is stored in R2 and a durable preprocessing job is created in Postgres. Workers claim the job's stages and persist intermediate outputs. This keeps expensive layout, OCR, captioning, and embedding work outside the edge request lifetime. Chat can then retrieve the processed representation without re-parsing the entire collection on every question.

TABLE 02 / Plane / Owns / Does not own

Plane	Owns	Does not own
Edge application	User experience, authenticated API routes, library and team workflows	Long-running GPU inference
Compute service	Document processing, retrieval, reasoning, artifact generation	The browser's local presentation state
Shared state	Tenant records, job state, embeddings, source and derived artifacts	Decisions about the truth of a generated answer

PART II / THE SYSTEM

03 / A document is more than extracted text.

Flattening a PDF into one text string discards the structure that often carries its meaning. A heading changes the interpretation of a passage; a table encodes relationships; a figure can be the only evidence for a result. Synapse turns source files into a structured intermediate representation before retrieval.

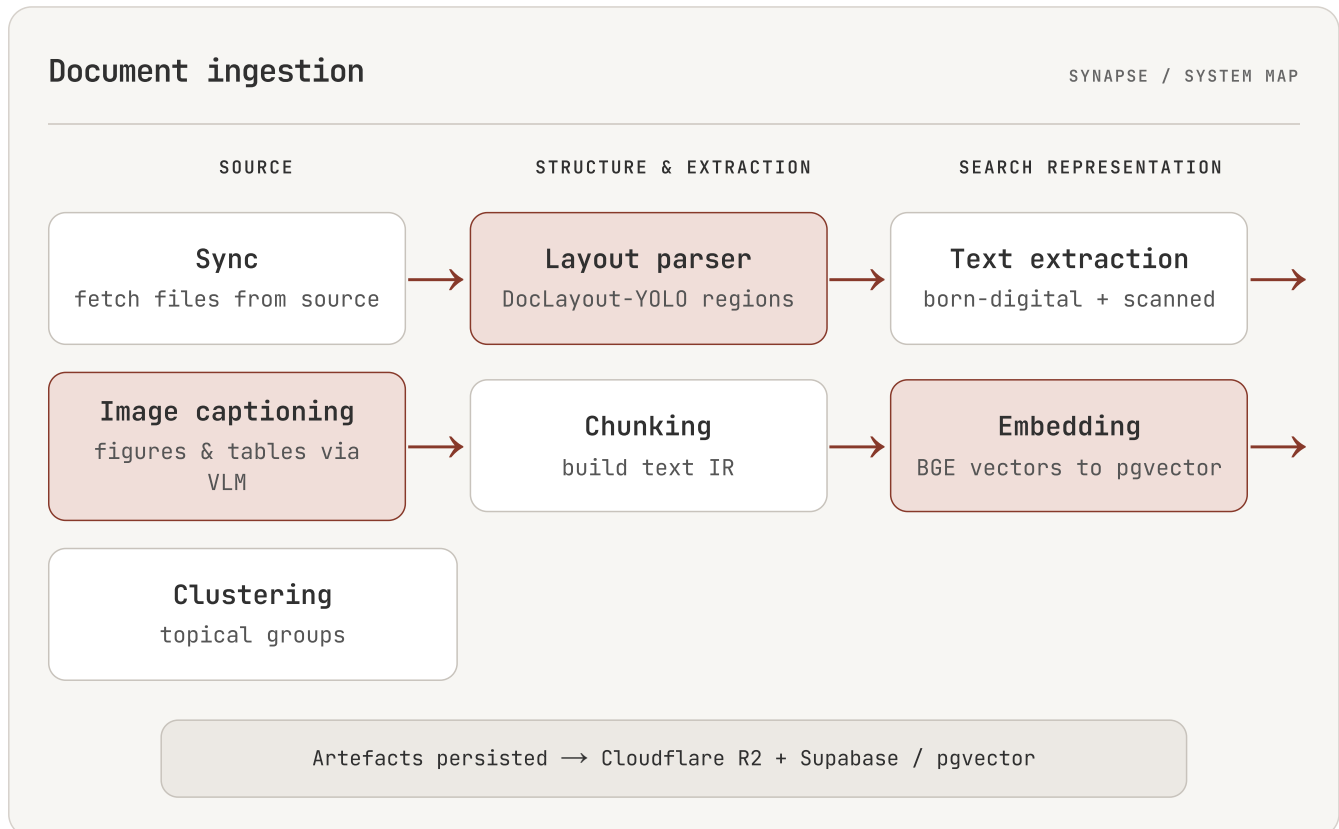


Figure 03. Queue-driven stages persist text, layout, visual artefacts, embeddings, and clusters as work advances.

TABLE 03 / Stage / Responsibility / Retained representation

Stage	Responsibility	Retained representation
Sync	Fetch source files and prepare page inputs	Document identity and source artifacts
Layout	Detect regions with DocLayout-YOLO	Paragraph, heading, table, figure, and other region boundaries
Text extraction	Read born-digital text and OCR scanned content	Text associated with document/page context
Image captioning	Describe or transcribe visual regions using a VLM	Captions, visual artifacts, and source links
Chunking	Construct retrieval-sized pieces with contextual and visual links	Text chunks with attribution
Embedding	Encode text and persist vectors	pgvector records linked to the underlying chunks
Clustering	Organize related content into topical groups	Semantic groupings over the processed library

The diagram presents the logical stage order. Extraction and captioning have separate worker paths after layout; their outputs are brought into the representation used for chunking. Individual stage processes let the worker planner balance model-heavy tasks against the host's available VRAM rather than loading every model in every process.

RECOVERABILITY

Queue-backed stages and persisted artifacts support resuming work after interruption. This is an architectural mechanism, not a claim that every failure is automatically recoverable or that the platform has a measured availability SLA.

PART III / EXECUTION

04 / Specialists for structure; configuration for reasoning.

The model roster separates document perception from text reasoning. A layout detector locates regions, an OCR system reads text, a vision-language model handles visual content, and retrieval uses a dense encoder with an optional cross-encoder reranker. Reasoning roles can select different model identifiers through configuration.

TABLE 04 / Role / Model / configuration / Implementation note

Role	Model / configuration	Implementation note
Layout detection	DocLayout-YOLO / DocStructBench weights	Region-aware document parsing
OCR	Surya; optional GLM-OCR fallback	Scanned-text extraction alongside born-digital parsing
Remote VLM	qwen/qwen2.5-vl-72b-instruct	Default caption API identifier through OpenRouter
Local VLM	Qwen/Qwen2-VL-2B-Instruct	Local configuration path when used
Embedding	BAAI/bge-large-en-v1.5	Default dense encoder; 1024-dimensional vectors
Inline reranking	BAAI/bge-reranker-v2-m3	Gated by retrieval configuration
Worker reranking	BAAI/bge-reranker-base	Worker-mode default; configurable separately
Reasoning	CHAT_GPT_MODEL plus per-role overrides	Planner, extractor, critic, grader, verifier, and synthesis configuration

DEFAULTS ARE NOT DEPLOYMENT CLAIMS

These identifiers describe the reviewed code and project configuration. The evaluation guide recommends bge-m3 for its benchmark setup. Results must be paired with their actual encoder, reranker, thinking mode, dataset slice, and run configuration; a default in the code is not proof of what ran in a particular deployment.

PART III / EXECUTION

05 / Retrieval follows the question.

A precise lookup and a comparison across many documents need different retrieval behavior. The planner classifies the request, rewrites it when useful, and decomposes complex questions into sub-queries. Spotlight questions can remain narrow; comparison, aggregation, multi-hop, and multi-entity questions may need broader evidence and follow-up searches.

The runtime fans out retrieval, combines dense and lexical candidates where enabled, expands neighboring chunks, and applies cross-encoder reranking. Retrieved material is distilled into an evidence brief while preserving attribution. A relevance/sufficiency grader and a completeness critic can identify gaps before synthesis. These checks are bounded by the selected reasoning mode.

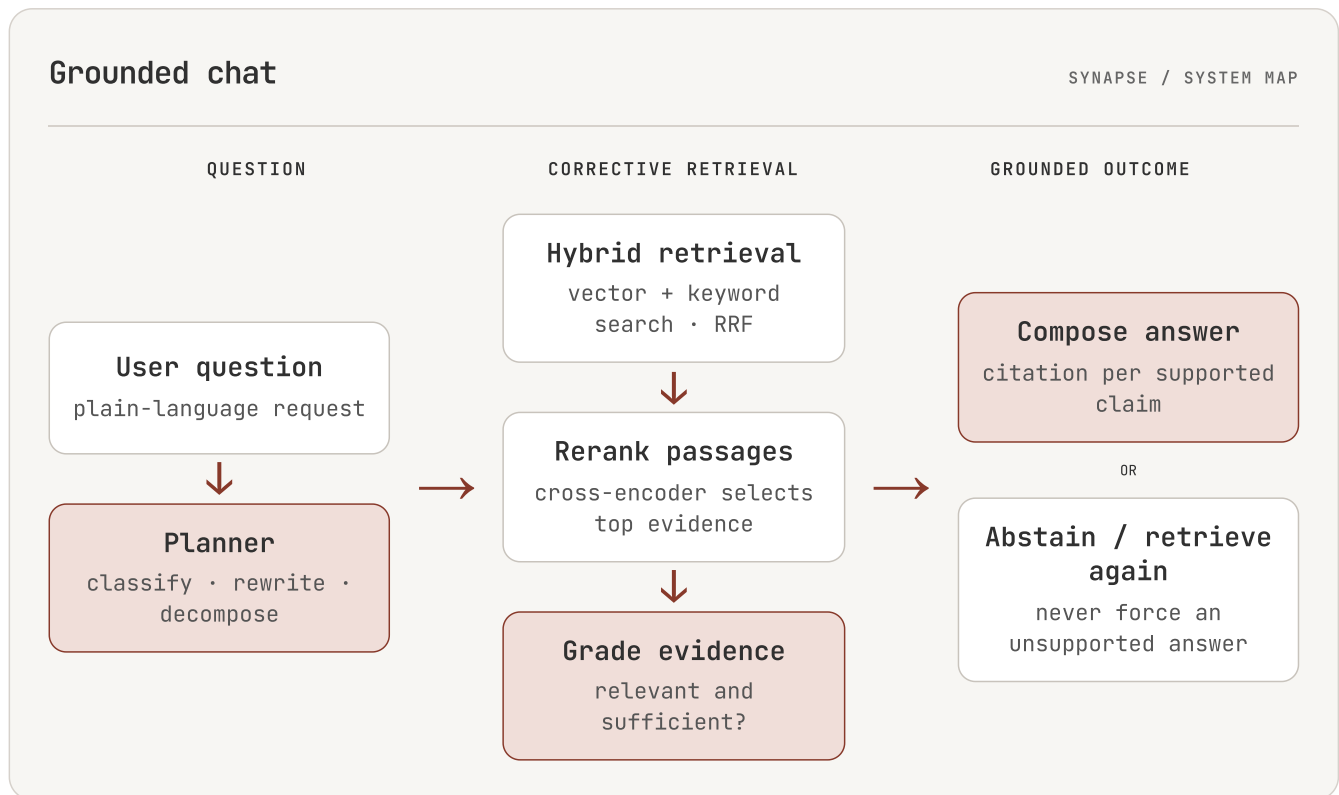


Figure 04. Weak evidence loops back through retrieval; sufficient evidence advances to a cited answer.

TABLE 05 / Role / Decision or output

Role	Decision or output
Planner	Intent, query class, rewritten query, and decomposition
Retriever	Ranked source passages and associated evidence
Extractor	A compact, source-anchored evidence brief
Retrieval grader	Whether the evidence is relevant and sufficient
Gap finder / completeness critic	Missing entities, attributes, or bridging questions
Synthesizer	An answer grounded in the consolidated evidence
Faithfulness verifier	Unsupported claims and a possible correction pass

Depth is a budget, not a promise of quality

TABLE 06 / UI mode / API value / Critic rounds / Sub-queries / Breadth / Answer token cap

UI mode / API value	Critic rounds	Sub-queries	Breadth	Answer token cap
Fast / low	0	0	1.0x	1600
Balanced / medium	1	2	1.25x	3200
Deep / high	2	4	1.5x	5500

The runtime can lower these limits with environment caps. More searches or critic calls consume latency and model budget; they do not guarantee a better answer. The useful question is whether a deeper mode recovers missing evidence on a defined evaluation slice.

AN IMPORTANT LIMIT

The reviewed faithfulness verifier has a fail-open fallback when its call fails. A verifier outage therefore must not be interpreted as a successful grounding check. Stronger degraded-status reporting and stricter policies for high-stakes use are sensible hardening work, not features claimed as already proven here.

PART III / EXECUTION

06 / A citation is part of the answer contract.

The synthesizer emits references that the server resolves to real source records. Citation and visual markers connect the narrative to retrieved chunks and visual artifacts; the client renders them as inspectable source chips and figure cards. This makes the evidence path available to the reader instead of hiding it behind a generic filename.

Evidence-linked answer representation

Question

- > retrieved chunks + document/page identity
- > distilled evidence brief
- > answer with `[[CITE:n]]` and `[[VISUAL:id]]` markers
- > resolved source records and visual artifacts
- > inspectable citation chips and inline figures

Source attribution and correctness are different measurements. A cited passage may be real but irrelevant, or a sentence may overstate what the passage establishes. That is why retrieval relevance, answer correctness, citation quality, and honesty are evaluated separately rather than combined into a single unqualified 'accuracy' number.

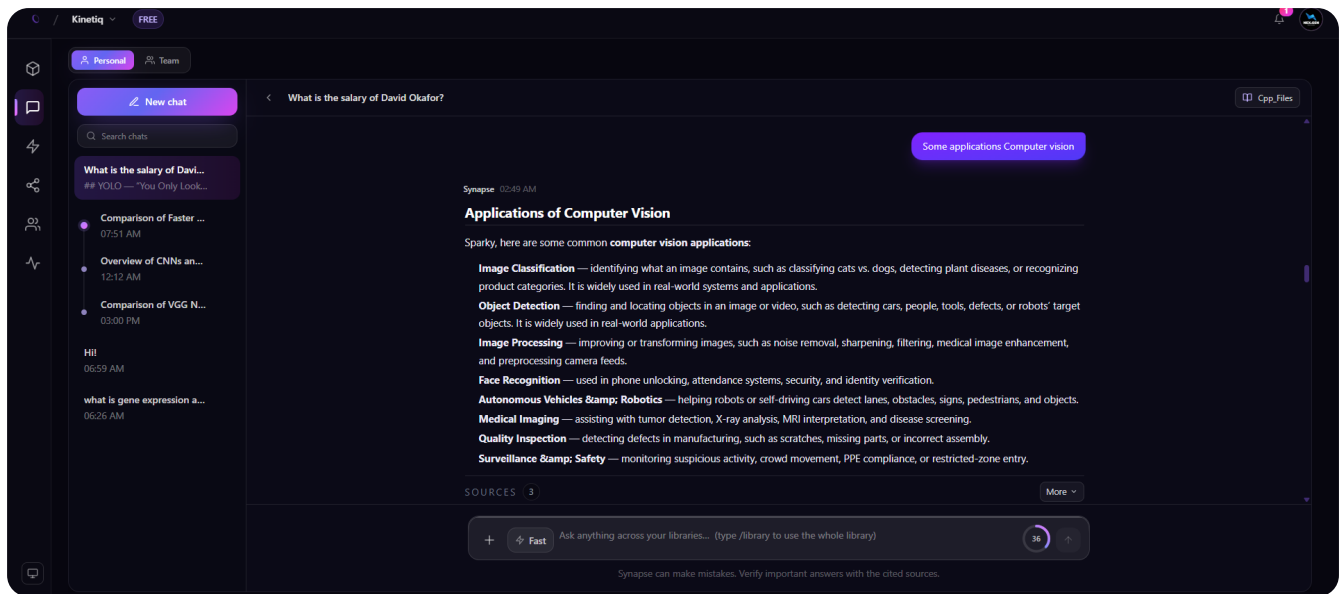


Figure 05. The answer surface keeps sources alongside the response so the reader can inspect the supporting material.

PART III / EXECUTION

07 / From an answer to a reusable artifact.

Agent mode handles requests whose output is not only prose. It plans the task, acquires data through parsing and retrieval, chooses an artifact form, and builds a Vega-Lite or Mermaid specification. Validation and a bounded repair path sit between specification generation and delivery.

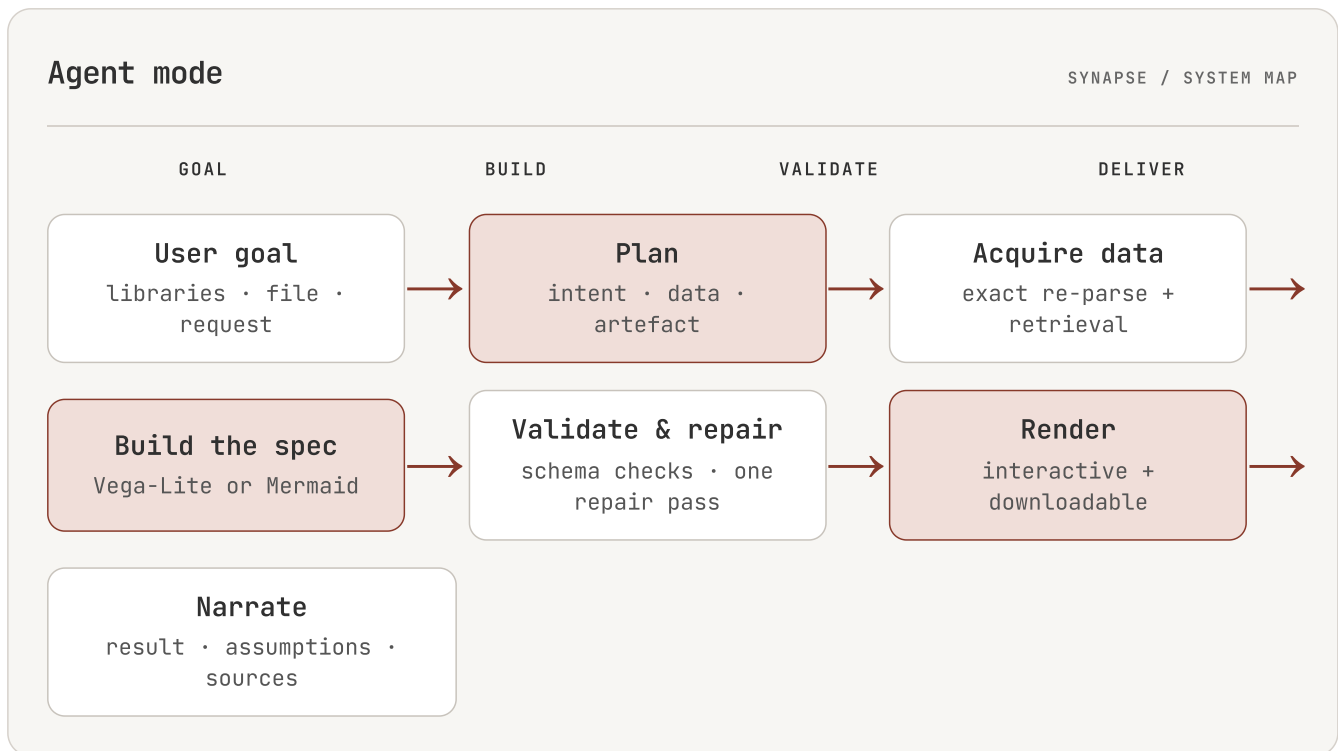


Figure 06. Data values are bound in code rather than authored by the model, keeping visuals tied to source evidence.

Numerical data is bound by code rather than left for the model to invent. The generated specification describes how the data should be presented; it is not permission to fabricate the underlying values. A narrative explains the result, assumptions, and sources after the artifact is produced.

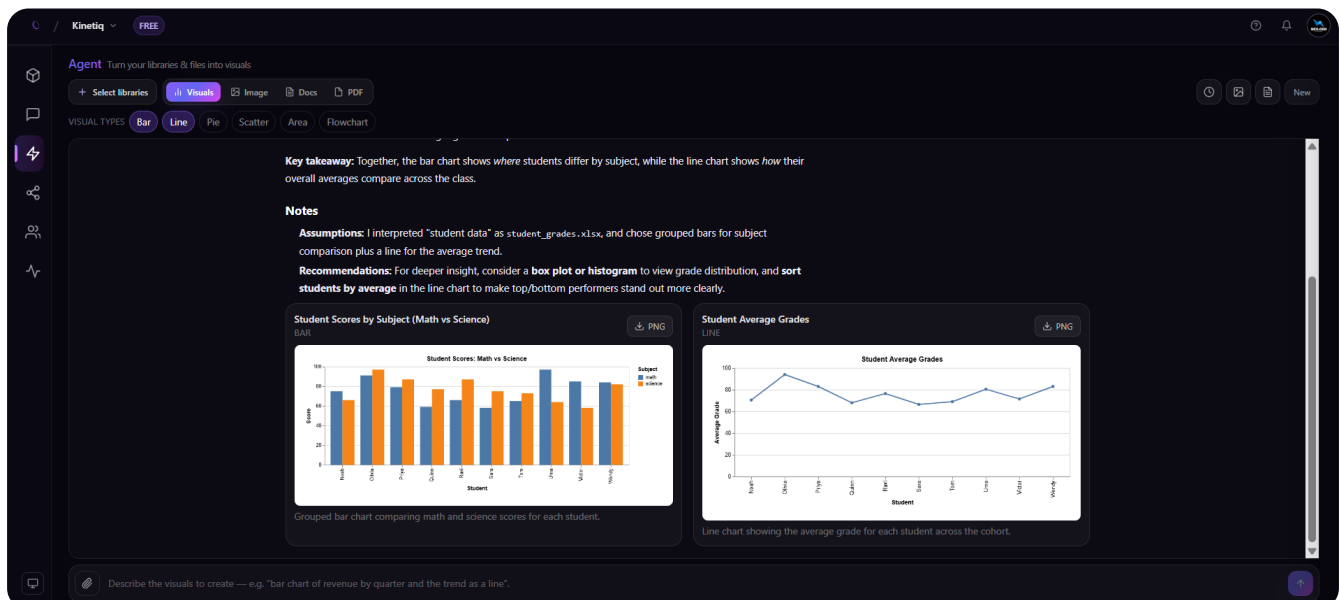


Figure 07. Visual outputs are delivered inside the workspace rather than as disconnected model text.

KEEP TWO TESTS SEPARATE

A specification can be syntactically valid while displaying the wrong data. Validation needs both a structural check and a source-value check; successful rendering alone does not establish numerical correctness.

PART III / EXECUTION

08 / Relationships retain their provenance.

The graph pipeline starts with processed library content, extracts entities and relationships, resolves duplicate names, and builds nodes and edges with confidence and source references. The resulting graph is persisted and served to the interface for exploration.

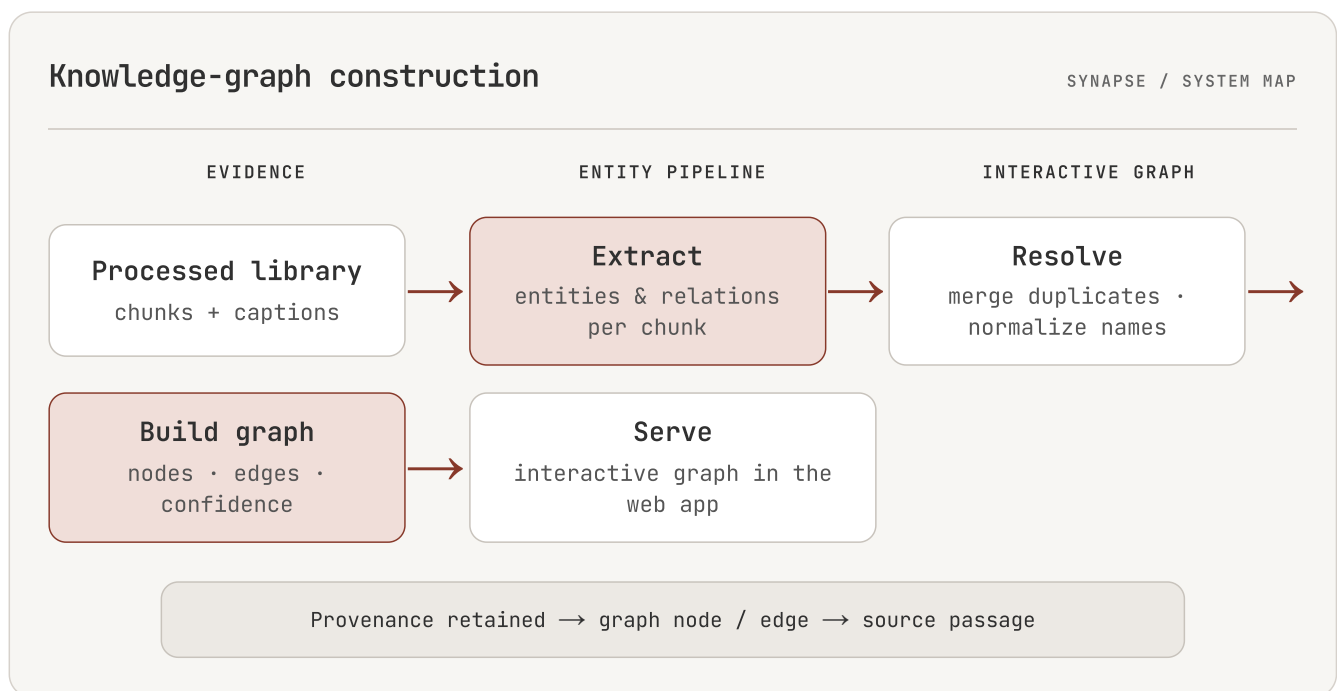


Figure 08. Every node and edge retains provenance back to the source chunk stored with the graph in Supabase.

A graph is another representation of the evidence, not an independent source of truth. Entity resolution can merge distinct subjects; extraction can infer a relationship that a passage does not actually state. Retaining the originating chunk allows a user to inspect a graph claim and provides a basis for targeted evaluation of extraction and resolution quality.

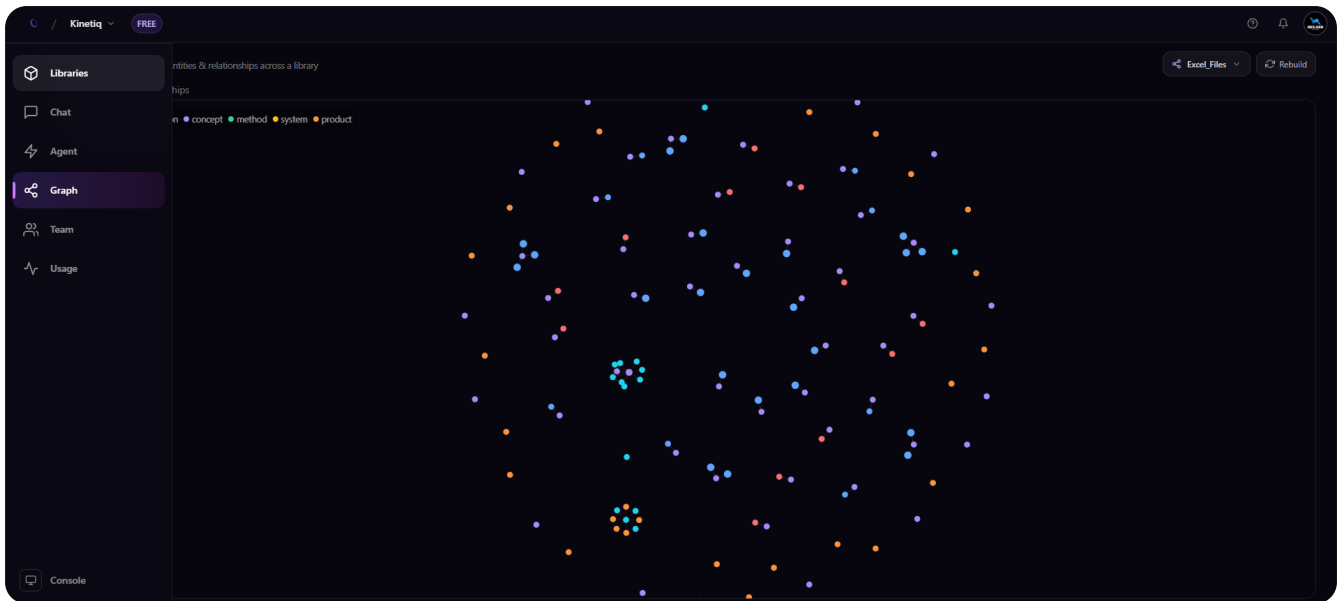


Figure 09. Graph exploration offers a relationship-oriented view of the processed collection.

PART IV / PLATFORM BOUNDARIES

09 / Collaboration does not mean unrestricted access.

Organizations, members, roles, libraries, conversations, and sharing relationships are modeled explicitly. Personal chats remain private, while shared team conversations can use libraries from participating organizations only through the sharing workflow. Postgres row-level security is part of the data boundary.

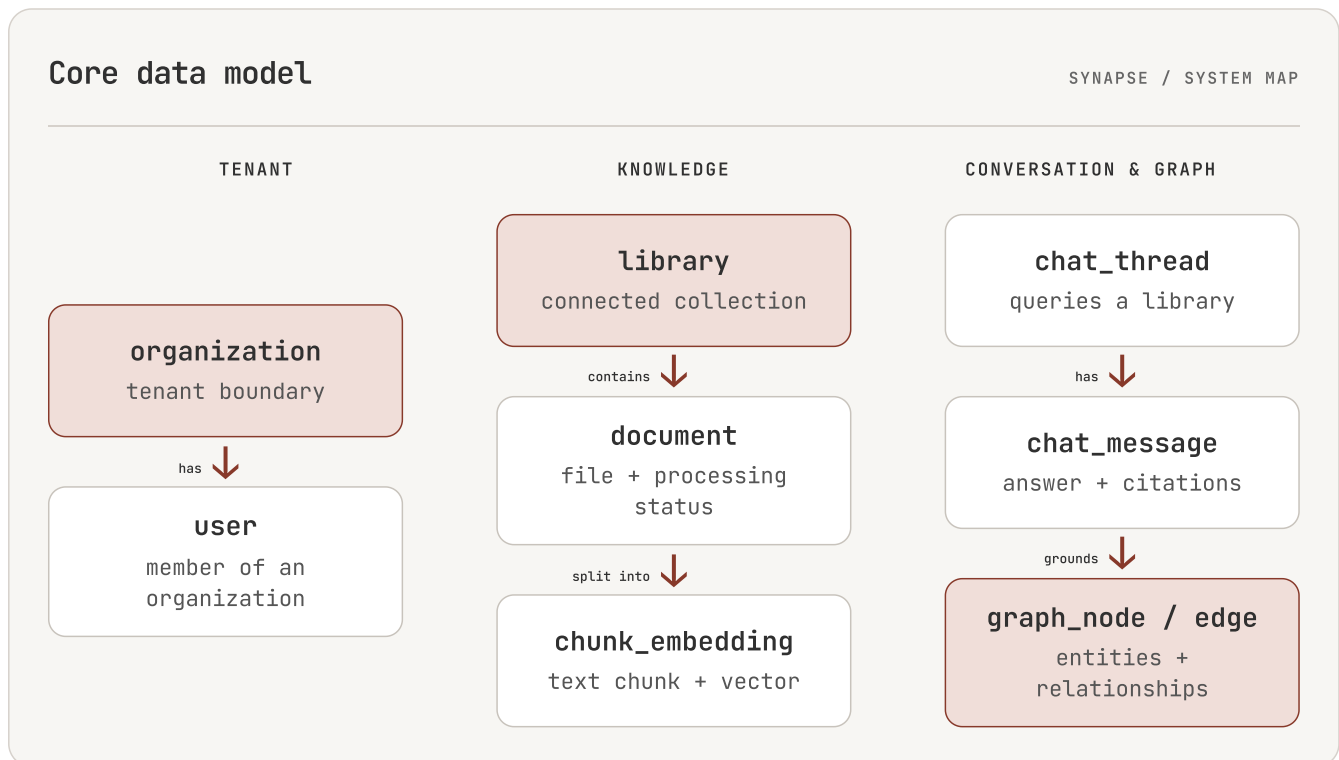


Figure 10. Rows are scoped to an organization and protected with row-level security.

Every access path still needs to enforce the caller's permissions: listing files, retrieving chunks, resolving citation previews, downloading artifacts, and accessing graphs. In particular, server-side service-role access must be paired with application authorization; database RLS by itself does not prove a service-role request is correctly scoped.

- Treat library sharing as an explicit grant, not an implicit consequence of joining a conversation.
- Test organization isolation on retrieval and artifact endpoints as well as the UI.
- Retain document and chunk identity when evidence moves between agents.
- Do not publish secret keys or infer security certification from the presence of an RLS policy.

SCOPE OF THIS STUDY

The repository includes tenancy and collaboration mechanisms; its roadmap also calls out permissions hardening. This document describes the architecture without claiming a completed independent security audit.

PART V / EVIDENCE AND EVALUATION

10 / Measure retrieval before judging the answer.

The evaluation harness uses DOUBLE-BENCH-oriented methodology with single-hop and multi-hop questions. Its full ingestion path converts sampled page images into PDFs and runs the actual parsing, captioning, chunking, embedding, and clustering pipeline. A cheaper OCR-text path starts at chunking and must be reported as a different ingestion condition.

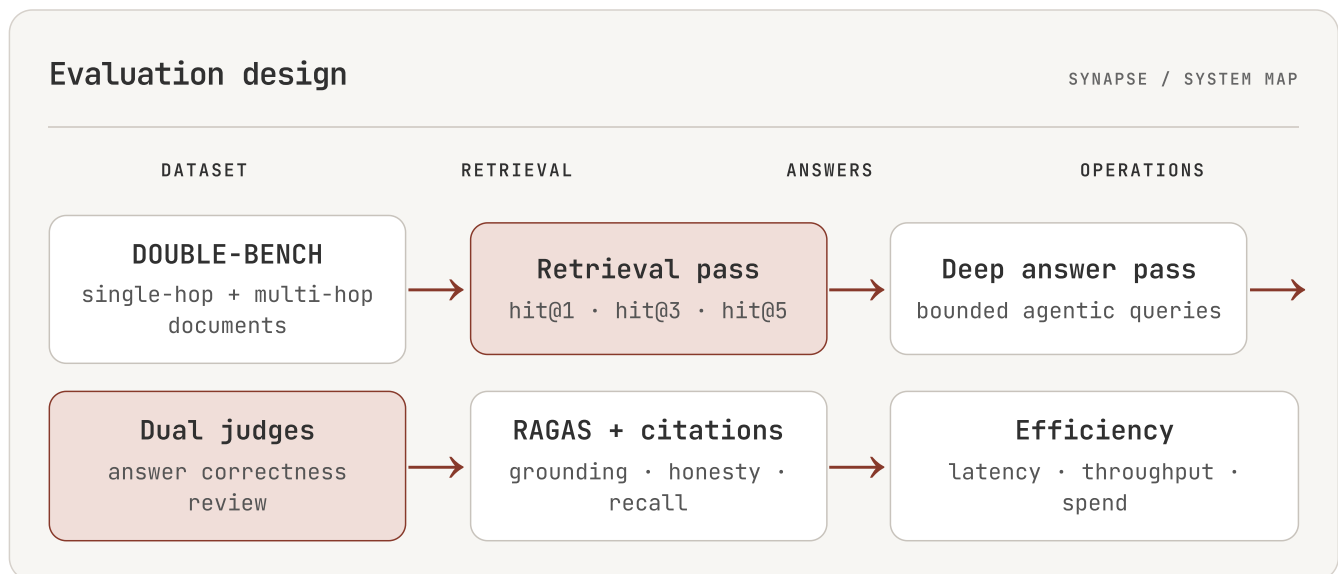


Figure 11. The harness defines the dataset, retrieval, answer-quality, grounding, and operational measurements shown in the result snapshots below.

Retrieval-only runs test whether the expected evidence is found without paying for answer synthesis and judging. Answer runs add dual judges, RAGAS-related measurements, citation checks, honesty/overconfidence, latency, throughput, and spend accounting. Persisted run records support aggregation and resuming work.

Page-level and document-level hits are not interchangeable

Rebuilt text PDFs may not preserve the benchmark's original page indices. A document-level match asks whether a retrieved passage belongs to the correct document. A page-level match asks a stricter question about the expected evidence page. The supplied graph below is explicitly document-level hit@5, not page-level retrieval accuracy.

Document-level hit@5

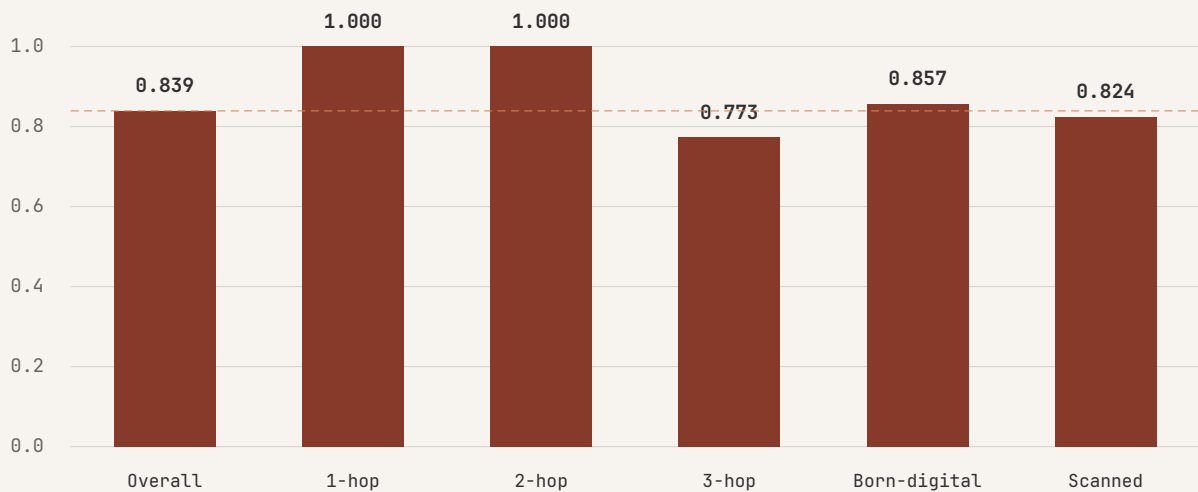


Figure 12. Retrieval results. Overall 0.839; 1-hop 1.000; 2-hop 1.000; 3-hop 0.773; born-digital 0.857; scanned 0.824. The dashed reference marks overall hit@5.

Dual-judge answer distribution

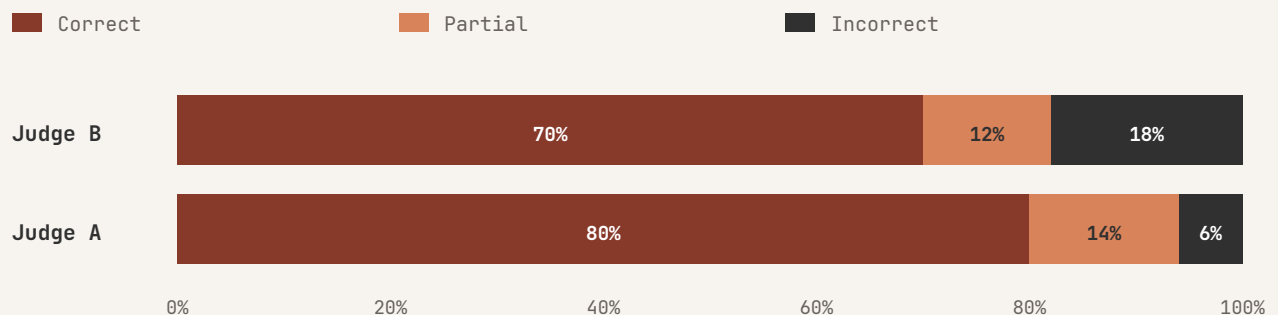


Figure 13. Answer quality. Judge A: 80% correct / 14% partial / 6% incorrect. Judge B: 70% correct / 12% partial / 18% incorrect.

RESULT PROVENANCE AND LIMITS

These values reproduce the supplied portfolio benchmark snapshots: overall document hit@5 = 0.839; 1-hop = 1.000; 2-hop = 1.000; 3-hop = 0.773; born-digital = 0.857; scanned = 0.824. Judge A reports 80% correct / 14% partial / 6% incorrect, and Judge B reports 70% / 12% / 18%. The snapshots do not establish the judged sample count, confidence intervals, or a full-suite result. No new evaluation run was performed for this document.

TABLE 07 / Measurement / What it establishes / What it does not establish

Measurement	What it establishes	What it does not establish
Document hit@5	The correct source document appeared among the retrieved candidates	The exact page was found, or the answer is correct
Judge distribution	The supplied judgments for the tested answer slice	Universal reliability or an independently verified ground truth
Citation checks	Agreement between claims, references, and supporting evidence under the metric	A security guarantee or complete factual verification

PART VI / OPERATING THE SYSTEM

11 / The job queue is part of the product.

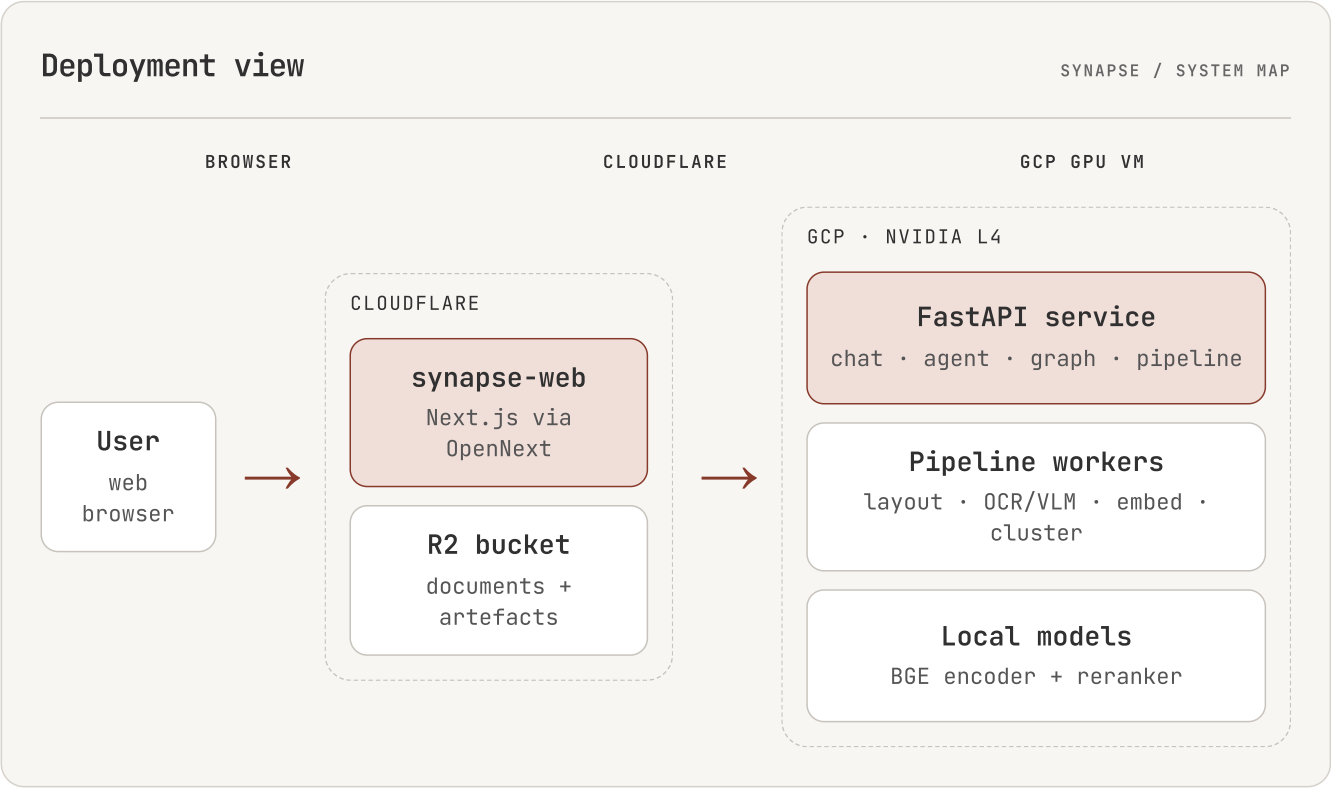


Figure 14. The edge application talks to a dedicated GPU service while R2 and Supabase provide durable shared state.

The documented deployment uses Cloudflare for the edge application, R2 for artifacts, Supabase for shared state, and a dedicated GPU host such as an NVIDIA L4 VM. Worker planning responds to the available hardware and queued stages. API-only operation is possible when GPU workers are disabled.

TABLE 08 / Operational surface / Purpose

Operational surface	Purpose
GET /health	Liveness of the API service
GET /ready	Configuration and GPU readiness checks
GET /hardware	Hardware information and worker planning visibility
Processing jobs and stage records	Durable progress and recovery context
Model and thinking-mode configuration	Explicit cost, depth, and latency controls

Useful monitoring separates queue delay, stage execution time, model/API failures, retrieval behavior, and answer-generation latency. Model names, prompt versions, dataset identity, and run settings should accompany evaluation reports. Otherwise a score change cannot be attributed to a particular code, data, or model change.

Repository responsibility map

TABLE 09 / Modules / Responsibility

Modules	Responsibility
app / pipeline_api / worker_bootstrap	Service composition, pipeline control, and worker lifecycle
sync_worker / layout_worker / extraction_worker / caption_worker	Document acquisition and perception
chunk_worker / embed_worker / cluster_worker	Searchable representations
chat_api / chat_agents / chat_runtime / chat_queue	Question orchestration, role decisions, retrieval, and queued work
agent_api / agent_agents / agent_data / agent_specs	Artifact planning, data acquisition, and specification construction
kg_extract / kg_resolve / kg_build / kg_api	Knowledge-graph construction and serving
eval	Dataset preparation, query runs, metrics, and reports

CLOSING / ENGINEERING JUDGMENT

12 / A system that exposes its assumptions.

The useful engineering lesson is not that more agents make a better product. It is that each responsibility has an explicit place: perception produces structure, retrieval produces candidate evidence, reasoning identifies gaps, synthesis produces a narrative, and evaluation tests those claims independently.

- Preserve structure before building search: tables and figures are evidence, not decoration.
- Bound autonomous work with runtime limits and observable stages.
- Separate retrieval success, citation correctness, and answer quality.
- Treat graphs and generated artifacts as derived views that still need provenance.
- Report uncertainty, degraded checks, and benchmark scope instead of turning a demo into an unqualified reliability claim.